

INTRODUCTION TO TIME SERIES ANALYSIS & FORECASTING :

A Python Implementation

II. Decomposition Methodology

The first approach to determining the presence of the various components in a time series is naturally by proceeding to the examination of the graphical representation of the series. Each component is hence expected to have a specific influence of the graphical aspect of the series. But this depth of analysis will not suffice in accurately providing answers to our various questions.

A. Detecting, Extracting and Forecasting the Seasonal component

1. Component detection methods

a. A tabular method: the Buys-Ballot and Sorted Buys-Ballot table

The first approach to the detection of the seasonal component is via the analysis of the Buys-Ballot table and its sorted version.

[Quick exercise of BB table and Sorted BB table in class]

The sorted BB table will indicate the persistence of a given season as the low (resp. highest) periodic values of the series. This allows us to clearly identify rigid seasonal patterns or lack thereof.

However, seasonal patterns can also be of random nature, *i.e.* the peaks and troughs might not always occur at the same season from one period to the other. This is what is known as random seasonality.

Q? : in the most extreme case of random seasonality, what would the sorted BB look like?

Hence the need for more accurate and precise tools.

b. Variance analysis test

The Fisher test stemming from the variance analysis of the BB table will provide alternate tools that account for the shortcomings previously discussed.

Seasonal and Periodic Averages

Seasons \ 1 ... s ... S	Periodic Averages
Periods	
1 $X_{1,1}$... $X_{1,s}$... $X_{1,S}$	$X_{p,\circ} = \frac{1}{S} \sum_{s=1}^S X_{p,s}$
...	
p $X_{p,1}$... $X_{p,s}$... $X_{p,S}$	
...	
P $X_{P,1}$... $X_{P,s}$... $X_{P,S}$	$X_{\circ\circ} = \frac{1}{PS} \sum_{p=1}^P \sum_{s=1}^S X_{p,s}$
Seasonal Averages	
$X_{\circ,s} = \frac{1}{P} \sum_{p=1}^P X_{p,s}$	

Variance Analysis Table

Sum of Squares	df	Denomination of Variance	Variance
$S_s = P * \sum_s (X_{o,s} - X_{oo})^2$	$S - 1$	Seasonal	$V_s = \frac{S_s}{S - 1}$
$S_p = S * \sum_p (X_{p,o} - X_{oo})^2$	$P - 1$	Periodic	$V_p = \frac{S_p}{P - 1}$
$S_R = \sum_p \sum_p (X_{p,s} - X_{p,o} - X_{o,s} + X_{oo})^2$	$(P - 1) * (S - 1)$	Residual	
$S_T = S_s + S_p + S_R$	$P * (N - 1)$	Total	

For each of these values, a Fisher test of influence is carried out to determine if the data is significantly influenced by season (resp. periodic) effects.

Seasonal influence (columns effect) test:

$$H_0 : \text{No influence}$$

Compute $F_c = \frac{V_s}{V_R}$ and compare it to a $\mathcal{F}_{v_1; v_2}$ Fisher value where $v_1 = s - 1$ and $v_2 = (P - 1) * (S - 1)$

If $F_c > \mathcal{F}_{v_1; v_2}$ then H_0 is rejected => presence of seasonal Component

a. Autocorrelation Function and Test **** optional

2. Seasonal Adjustment, Extraction and Seasonal forecast methods

Once it has been established that the series contains a seasonal component, the next step is to remove said seasonality from the series, without affecting the remaining components, in order to be able to study them. This process is called seasonal adjustment of the series and is carried out by computing seasonal coefficients of the series.

It goes without saying that the seasonal adjustment of the series is subject to the decomposition model. But in a nutshell, the seasonal coeffs are obtained by computing seasonal averages over the lifespan of the series.

Hence for a given series, we'll have $\text{Seasonal coeffs} = S' = \frac{1}{S} \sum_{j=1 \dots S}$

i.e. for the first season (month, or week, or quarter), a first coefficient s_1 is obtained as:

$s_1 = \text{average}(X_{1,1}; X_{2,1}; \dots; X_{P,1})$. Then we also have

$s_2 = \text{average}(X_{1,2}; X_{2,2}; \dots; X_{P,2})$ etc.

For reasons clarified supra, these coefficients will be referred to as provisional or temporary seasonal coefficients.

a. The conservation principle

Analysis on the seasonal component is aimed at revealing a new distribution of the intra-annual profile of the series, without modifying the annual cumulated value reached. In other words, the annual averages of the raw series and the seasonally adjusted series are supposed to be identical. This basic principle is known as the conservation principle, and it comes in handy when we want to compute the real seasonal coefficients of the series.

- In the case of a series assembled in an additive manner, the seasonal coefficients are supposed to sum up to zero ($S = \sum_{j=1}^S s_j = 0$)

If $S = 0$ then the previous temporary coefficients will be taken as final seasonal coeffs.

If $S \neq 0$ then they will need to be normalized to obtain the final coeffs as follows $S_j^* = s_j - \bar{S}$

- In the framework of a multiplicative model, the condition to be met by the coeffs is that their average =1.

$\bar{S} = \frac{1}{S} \sum_{j=1}^S s_j$ If $\bar{S} \neq 1$ then they will need to be normalized to obtain the final coeffs as follows $S_j^* = s_j / \bar{S}$

b. Seasonal forecasts